LESSON PLAN

PROPELLER THEORY (7)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	01
Total No. of Lesson Plans	8	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Introduction, propeller function, tractor and pusher types, material need and construction of propeller
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Introduction, propeller function, tractor and pusher types, material need and construction of propeller
5. Specific Objectives	7.1.1.1 Trainee will be able to describe the various types of propellers, principles of operation and pitch change mechanisms.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of propeller diagrams; Safety: Do not touch rotating parts; Supervision: Instructor monitors diagram interpretation
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a propeller?
(b) Name two common materials used in propeller construction.
(c) Define the term 'tractor configuration'.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Propeller function and thrust generation • Tractor and pusher configuration types • Blade material requirements and selection • Basic construction components and assembly	L & Ds	PPT, IB	15 min
(b)	• Principles of operation and aerodynamic forces • Fixed pitch versus variable pitch mechanisms • Feathering and reversing operational concepts • Structural integrity and stress distribution	L & Ds	PPT, PDF	20 min
(c)	• Maintenance implications of different materials • Construction tolerances and quality standards • Integration with engine power absorption • Summary of propeller system types	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the difference between a tractor and a pusher propeller installation.
(b) Describe how a variable pitch mechanism adjusts blade angle during flight.

12. Home Assignment and Follow Up	Prepare a brief note on power absorption characteristics to be discussed in Lesson Plan 02 regarding feathering and braking advantages.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Definition of terms, power absorption, feathering & braking & its advantages.	Refer to the master syllabus reference list	LP 02

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min